

IT 620 Wireless Networks Security and Administration

Instructor Ryan Tolboom

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Office 3506 Guttenberg Information Technologies Center (GITC)

Office Hours 12:00PM to 1:00PM on Thursdays and Fridays or via Zoom (email to schedule)



Objective

This course introduces the fundamentals of wireless network security and administration. Topics include: wireless LAN vulnerabilities, passive and active wireless attacks, enterprise wireless hardware security, secure wireless authentication and communication, wireless intrusion detection and prevention systems, WiFi and cellular network management, location privacy, personal area network administration and security, mobile IP security, GSM, CDPD, 3G and 4G network security. The course provides both a theoretical foundation and hands-on experience in these areas.

Grading

- 20% Labs
- 20% Quizzes
- 20% Project
- 20% Midterm Exam
- 20% Final Exam

Course Materials

- A laptop meeting the [YWCC minimum specs](#)
- [VirtualBox](#) installed and working
- [git](#) installed and working
- [Python](#) installed and working
- [Mininet-WiFi](#)
- Textbook: [Wireless Networking in the Developing World \(WNDW\)](#)
- Textbook: [High Performance Browser Networking \(HPBN\)](#)

Learning Outcomes

1. Physical

- 1.1 Identify physical RF infrastructure
- 1.2 Explain how the properties of electromagnetic waves can be used to transmit information
- 1.3 Recognize the unique problems of RF propagation

2. WiFi

- 2.1 Monitor and analyze WiFi traffic
- 2.2 Create a managed WiFi network
- 2.3 Perform and analyze a deauth attack on a WiFi network

Course Outline

Week	Presentations	Exercises	Learning Outcomes
1			
2			
3			
4			
5			
6			
7	• Midterm		
8			
9			
10			
11			
12			
13			
14			

Week	Presentations	Exercises	Learning Outcomes
15			
Finals	• Final		

Academic Integrity

Academic Integrity is the cornerstone of higher education and is central to the ideals of this course and the university. Cheating is strictly prohibited and devalues the degree that you are working on. As a member of the NJIT community, it is your responsibility to protect your educational investment by knowing and following the academic code of integrity policy that is found at: [NJIT Academic Integrity Code](#) .

Please note that it is my professional obligation and responsibility to report any academic misconduct to the Dean of Students Office. Any student found in violation of the code by cheating, plagiarizing or using any online software inappropriately will result in disciplinary action. This may include a failing grade of F, and/or suspension or dismissal from the university. If you have any questions about the code of Academic Integrity, please contact the Dean of Students Office at dos@njit.edu.

Generative AI

Student use of artificial intelligence (AI) is permitted in this course for non-exam assignments and activities. Additionally, if and when students use AI in this course, the AI must be cited as is shown within the [NJIT Library AI citation page](#) for AI. If you have any questions or concerns about AI technology use in this class, please reach out to your instructor prior to submitting any assignments.

Student Absences for Religious Observations

NJIT is committed to supporting students observing religious holidays. Students must notify their instructors in writing of any conflicts between course requirements and religious observances, ideally by the end of the second week of classes and no later than two weeks before the anticipated absence. All instructors must include a reminder on the course syllabus about this notification process. All instructors are required to provide academically reasonable accommodations, allowing students to complete missed assignments, exams, quizzes, or other coursework within the term. Instructors are encouraged to consider the [NJIT religious holiday calendar](#) and exercise cultural sensitivity when scheduling assessments or major assignments. All instructors must ensure that students are not penalized for properly documented absences and maintain confidentiality regarding religious observances. For questions or additional guidance, please [review the policy](#) or contact the Office of Inclusive Excellence at inclusiveexcellence@njit.edu.